Journal of Psychopathology and Clinical Science

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Online First Publication, July 9, 2026. https://dx.doi.org/10.1037/abn0001150

CITATION

Fisher, H., Jaffe, N., Pidvirny, K., Tierney, A. O., Pizzagalli, D. A., & Webb, C. A. (2026). Using natural language processing to track negative emotions in the daily lives of adolescents. *Journal of Psychopathology and Clinical Science*. Advance online publication. https://dx.doi.org/10.1037/abn0001150

AMERICAN PSYCHOLOGICAL ASSOCIATION

Journal of Psychopathology and Clinical Science

© 2026 American Psychological Association
ISSN: 2769-7541

https://doi.org/10.1037/abn0001150

Using Natural Language Processing to Track Negative Emotions in the Daily Lives of Adolescents

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Tracking emotion fluctuations in adolescents' daily lives is important for understanding mood dynamics and identifying early markers of affective disorders. This study examines whether natural language processing (NLP) methods, including large language models (LLMs), can approximate adolescents' concurrent self-reported negative affect from ecological momentary assessment narratives. We compared nomothetic (group-level) and idiographic (individualized) models as well as different NLP techniques for capturing within-person emotion fluctuations. Ecological momentary assessment text responses from 97 adolescents (ages 14–18, 77.3% female, 22.7% male, N_{EMA} = 7,680) were analyzed. Text features were extracted using dictionary-based approaches, topic modeling, and emotion ratings derived from a LLM (GPT-4o). Random Forest and Elastic Net Regression models predicted negative affect from text features, and SHapley Additive exPlanations values were used to assess feature importance. Key findings, interactive visualizations, and model comparisons are available via the website: https://emotracknlp.streamlit.app/. Idiographic models combining text features from different NLP approaches performed comparably to nomothetic models and to Generative Pre-Trained Transformer 4 Omni (GPT-4o) alone in R^2 (nomothetic: .06–.11, idiographic: .06–.10, GPT-4o: .06–.10) but yielded lower root-mean-squared error (nomothetic: .52–.88, idiographic: .47–.76, GPT-4o: .85–1.36), improving within-person precision. Importantly, there were substantial between-person differences in model performance and predictive linguistic features. When selecting the best-performing model for each participant (e.g., nomothetic or idiographic), significant correlations between predicted and observed emotion scores were found for 90.7%–94.8% of participants. Findings suggest that although nomothetic models and LLMs offer scalability, idiographic models provide greater predictive precision with sufficient within-person data. A personalized approach selecting optimal models for each individual may enhance emotion monitoring and inform targeted interventions through contextual text insights.

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Andrew Schwartz served as action editor.

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This research was supported by National Institute of Mental Health K23MH108752, the Tommy Fuss Fund, a NARSAD Young Investigator Grant from the Brain and Behavior Research Foundation, and the Klingenstein Third Generation Foundation (Christian A. Webb). Hadar Fisher was partially supported by an HMS Psychiatry Research Livingston Award. Christian A. Webb was partially supported by National Institute of Mental Health R01MH116969 and NCCIH R01AT011002. Diego A. Pizzagalli was partially funded by National Institute of Mental Health R37MH068376 and P50MH119467.

Christian A. Webb has received consulting fees from King and Spalding law firm. Over the past 3 years, Diego A. Pizzagalli has received consulting fees from Abbvie, Boehringer Ingelheim, Circular Genomics, Compass Pathways, Engrail Therapeutics, Neumora Therapeutics, Neurocrine Biosciences, N1 Biocorp., Neuroscience Software, Syntropic Medical Takeda, and Xenon Pharmaceuticals; he has received honoraria from the American Psychological Association, Psychonomic Society and Springer (for editorial work) and from Alkermes; he has received research funding from the Brain and Behavior Research Foundation, Circular Genomics, Dana Foundation, Wellcome Leap, Millennium Pharmaceuticals, and National Institute of Mental Health; he has received stock options from Ceretype Neuromedicine, Compass Pathways, Engrail Therapeutics, Neumora Therapeutics, and Neuroscience Software. No funding from these entities was used to support the current work, and all views expressed are solely those of the authors. The other authors declare no competing financial interests.

Hadar Fisher served as lead for conceptualization, formal analysis, methodology, visualization, and writing–original draft. Nigel Jaffe contributed equally to data curation and served in a supporting role for writing–review and editing. Kristina Pidvirny contributed equally to data curation and served in a supporting role for methodology and writing–review and editing. Anna O. Tierney contributed equally to data curation and served in a supporting role for writing–review and editing. Diego A. Pizzagalli contributed equally to writing–review and editing and served in a supporting role for supervision. Christian A. Webb served as lead for data curation, funding acquisition, project administration, resources, and supervision and contributed equally to writing–review and editing.

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General Scientific Summary

This study shows that natural language processing and machine learning methods can predict adolescents' negative emotional states from text responses with reasonable accuracy. Personalized models that adapt to each individual's language performed better than general models, highlighting the potential of tailored approaches for improving emotion monitoring.

Keywords: Natural language processing, negative affect, personalized monitoring

Supplemental materials: https://doi.org/10.1037/abn0001150.supp

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Emotions play a fundamental role in human life, serving as essential cues that influence our actions and interactions with the environment (Barrett et al., 2007). Emotional states act as immediate alerts to potential benefits or dangers, driving us toward actions that align with our personal goals and away from potential threats (Frijda, 1988). Research shows that not only the intensity of emotions but also their temporal dynamics can contribute to the development of mood disorders (Fisher, Fatimah, et al., 2025; Fisher, Pidvirny, et al., 2025; Houben & Kuppens, 2020; Kuppens & Verduyn, 2017). This is especially important during adolescence, a developmental period marked by increases in the frequency of negative emotional states, which may contribute to elevated risk for depression (Bailen et al., 2019; Bennik et al., 2014; Hollenstein & Lanteigne, 2018). Developing effective methods to assess daily emotional changes that are specifically suitable for adolescents is crucial. These measurements could identify both acute emotional distress and longer term maladaptive patterns in emotions, serving two key purposes: enabling timely interventions during periods of high negative affect and facilitating targeted intervention to address maladaptive emotional patterns.

Language offers a powerful avenue for measuring emotion, as people convey their emotional states both explicitly and implicitly through their choice of words and their manner of speaking or writing (Davitz, 2013; Pennebaker et al., 2003). Language plays a dual role in the realm of emotions, serving not only as a primary tool for expressing and communicating emotional states but also in actively shaping how emotions are experienced (Jablonka et al., 2012; Lindquist, 2017). Therefore, the current study aims to evaluate whether linguistic patterns in adolescent's descriptions of emotion-eliciting experiences can be used to track within-person fluctuation in emotional states.

With the rise of digital communication, people, especially adolescents, now express themselves extensively through text-based interactions, including social media posts, messages, and online discussions. This shift has generated an unprecedented volume of language data, offering a unique opportunity to study adolescents' emotions on a larger scale (Iliev et al., 2015; Mihalcea et al., 2024). Advances in natural language processing (NLP), a multidisciplinary field combining computer science, artificial intelligence, and linguistics, have enabled researchers to analyze these data efficiently and meaningfully (Hirschberg & Manning, 2015; Popel et al., 2020). Although NLP has been used in behavioral science for decades, its accessibility and affordability have significantly improved, making it an increasingly popular tool (Feuerriegel et al., 2025). One key advantage of NLP over traditional, manual text analysis is its ability to handle large data sets quickly and efficiently. By leveraging these advancements, NLP methods can identify patterns in language use (e.g., sentiment, tone, and self-referential quality), which can be used to approximate self-reported emotion ratings (Carlier et al., 2022; Kahn et al., 2007; Sun et al., 2020). Importantly, text-based communication provides insights not only into the intensity of emotions but also into the context in which those emotions arise. This contextual information can reveal situational factors contributing to emotional experiences, offering a more nuanced understanding of emotional processes.

In recent years, various text analysis tools have emerged (Boyd et al., 2022; Demszky et al., 2023; Eichstaedt et al., 2021; Neuendorf, 2017; Rathje et al., 2024; Tausczik & Pennebaker, 2010; van Loon, 2022), ranging from basic approaches like counting word frequencies to more advanced methods, such as leveraging large language models (LLMs), each with its own set of advantages and limitations. For example, closed-vocabulary programs (i.e., dictionary-based approach)[1] such as the Linguistic Inquiry and Word Count (LIWC; Pennebaker, 2001), use predefined dictionaries to categorize words. Although highly interpretable, transparent, and efficient at summarizing concepts, these methods often neglect context, leading to potential misinterpretations (Eichstaedt et al., 2021). LIWC may classify the word *mad* as an indicator of anger, yet in phrases like "She's mad talented," it conveys emphasis rather than negativity, illustrating how context-dependent meanings can lead to inaccurate classifications. In contrast, open-vocabulary approaches, such as Latent Dirichlet Allocation (LDA), and word embedding methods, leverage data-driven techniques to examine a broader spectrum of words and topics (Blei et al., 2003; Griffiths et al., 2007; Sivakumar et al., 2020). These methods are better at capturing nuances, addressing ambiguous word meanings, and are less susceptible to misinterpretations. Limitations of open-vocabulary approaches include the need for more technical expertise, larger data sets, and careful consideration of parameter choices, as well as challenges in interpretability (Eichstaedt et al., 2021; van Loon, 2022).

More recently, LLMs like Generative Pre-Trained Transformer (GPT) have shown great promise in accurately rating various psychological constructs in text, overcoming many of the constraints of older methods (Rathje et al., 2024). These models can interpret the context of words and have achieved effective results across multiple languages with simple prompts (Feuerriegel et al., 2025). The main limitations of this approach include a lack of transparency in

[1] We use the terms "closed-vocabulary" and "dictionary-based" interchangeably to refer to approaches that rely on predefined word categories (e.g., LIWC). These methods contrast with open-vocabulary approaches, which derive features directly from the text data without relying on preset dictionaries, often using machine learning or statistical models to identify relevant patterns.

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how inferences were generated and difficulties in reproducing results due to their probabilistic nature (Abdurahman et al., 2024). For example, although LLM-based approaches could predict whether a person experiences negative emotions based on social media posts, LLMs could not be used to identify which linguistic cues (such as the use of pronouns) predict this experience (Feuerriegel et al., 2025; Rathje et al., 2024). In the latter case, dictionary-based methods are preferrable. Ultimately, closed- and open-vocabulary approaches, along with advancements in LLMs, provide complementary strengths that may significantly enhance our ability to understand psychological states through language. By combining these methods, researchers may be able to leverage their unique advantages while mitigating their limitations, offering a robust alternative to traditional self-report measures assessing emotions.

Despite the increasing sophistication of these methods, significant gaps remain in the current literature. Most importantly, existing studies on text-based emotion prediction have focused on exploring between-person differences (nomothetic approach), primarily identifying which text features are associated with individuals experiencing high levels of negative emotions (e.g., Tackman et al., 2019). These approaches aggregate data across participants and aim to pinpoint common linguistic markers, such as the use of first-person pronouns or negative sentiment, that are indicative of heightened emotional distress and depression (Bathina et al., 2021; Funkhouser et al., 2024). However, this method overlooks both between-person differences in how linguistic markers relate to emotions and within-person fluctuations in emotional states. Even when using multilevel models to predict within-person fluctuations in emotions, models assumes that the same linguistic features signal high levels of negative emotions across different individuals (Funkhouser et al., 2024).

Such assumptions fail to account for the fact that language is inherently idiosyncratic, with individuals expressing emotions in unique and context-dependent ways. Beck and Jackson (2022) demonstrated the importance of idiographic approaches, which focus on person-specific patterns, by showing that the psychological and situational antecedents that predicted future loneliness varied substantially across participants, with no two individuals showing the same pattern of predictive features. This highlights the need for personalized models that adapt to the distinctive ways people express emotions through text, enabling more accurate predictions tailored to each individual. To address this limitation, this study integrates diverse language-based tools and leverages machine learning to develop personalized models that use written narratives to infer within-person fluctuations in self-reported negative affect.

The study's primary goal was to examine whether within-person fluctuations in negative emotions in adolescents' daily lives can be accurately tracked through text analysis. Because individuals express emotions in a variety of linguistic ways, this research aims to develop more personalized approaches to emotion tracking. Specifically, we address two key questions:

1. Do idiographic (individual-level) models outperform nomothetic (group-level) models in predicting within-person fluctuations in concurrent self-reported emotions?
2. Does combining various NLP approaches improve emotion prediction accuracy?

We compare three types of NLP approaches, which, as noted earlier, have complementary strengths and limitations:

1. Closed-vocabulary approach (dictionary-based approach: LIWC and Valence Aware Dictionary and sEntiment Reasoner [VADER])
2. Open vocabulary (LDA)
3. GPT-derived ratings

By addressing these questions, the study aims to benchmark the performance of different language-based approaches for predicting self-reported negative affect from ecological momentary assessment (EMA) narratives while comparing nomothetic and idiographic modeling strategies. This approach allows us to evaluate how different NLP methods perform in capturing within-person variation in self-reported affect and to assess their potential utility for modeling emotional experiences in daily life. In doing so, this work may inform future efforts to develop language-based approaches for monitoring emotional fluctuations and supporting scalable interventions for youth.

Method

Participants

Participants were derived from two larger studies that recruited adolescents with elevated levels of anhedonia, as well as typically developing (nonanhedonic) adolescents. They included 97 English-speaking adolescents aged 14–18 (75 female, 22 male; M_{age} = 16.2, SD = 1.9) recruited from the greater Boston area from January 2016 to November 2023. Participants were excluded based on a history or current diagnosis of any of the following *Diagnostic and Statistical Manual of Mental Disorders* (5th ed.) psychiatric illnesses: schizophrenia spectrum or other psychotic disorder, bipolar disorder, substance or alcohol use disorder within the past 12 months, or lifetime severe substance or alcohol use disorder. Participants were also excluded based on current diagnosis of anorexia nervosa or bulimia nervosa or if they had a neurodevelopmental disorder that would interfere with study tasks. Due to the neuroimaging component of this study, additional exclusion criteria included fMRI contraindications. For additional information about sample inclusion and exclusion criteria, see Murray et al. (2023).

Procedure

All procedures were approved by the Mass General Brigham IRB. Participants who were 18 years of age provided written informed consent; participants who were under 18 provided written assent, with their parents providing written consent. At the baseline session, either in-person or over Zoom, participants were administered a semistructured clinical interview, the Kiddie Schedule for Affective Disorders and Schizophrenia (K-SADS; Kaufman et al., 1997), and completed self-report measures. Following the baseline session, participants installed the MetricWire application on their smartphones to complete EMAs. Of the participants, 39 completed the EMA for four consecutive weeks; for the first 5 days, surveys were delivered 2–3 times per day in the afternoon and evening, and for the remaining 25 days, surveys were delivered four times per day. The other 58 participants completed the EMA in 5-day (Thursday through Monday) blocks every other week, with two to three survey prompts per day, for a median of 17 weeks.

Measures

Ecological Momentary Assessment

Participants were asked to rate on a 5-point Likert scale, ranging from 0, *very slightly or not at all*, to 4, *extremely* the extent to which they were feeling several emotions immediately before they started the assessment. Participants' negative affect (NA) included responses for: "sad," "nervous," and "angry." Mean NA was measured by averaging these three variables. In addition, participants responded to the following open-ended questions: (a) What were you thinking about right before you started this survey? (b) Think about the most enjoyable or happy time since you completed the last survey (or if this is your first survey, then the last 24 hr). Very briefly, what happened (—one to two sentences is fine)? (c) Think about the most stressful or negative time since you completed the last survey (or if this is your first survey, then the last 24 hr). Very briefly, what happened (—one to two sentences is fine)? Participants with fewer than 30 observations were excluded from the analysis.

Language Measures

We used the following strategies to extract text features and generate quantitative summaries of the language data:

Closed Vocabulary. For close vocabulary (dictionary-based approach), we extracted features using the LIWC (Tausczik & Pennebaker, 2010) and the VADER (Hutto & Gilbert, 2014). LIWC is a computerized text analysis tool that categorizes words into over 90 linguistic and psychological dimensions based on an internal dictionary of approximately 6,400 words (Boyd & Schwartz, 2021). It calculates the percentage of words that match each predefined category, offering insights into linguistic structures (e.g., pronouns), psychological constructs (e.g., affect), and broader language patterns (e.g., analytical thinking). LIWC has been extensively validated across numerous studies and is widely used in psychological research to quantify language use patterns associated with various psychological states and traits. For this study, we used LIWC-22 (Boyd et al., 2022), the latest version of the software, the analyze participants' responses to the EMA open questions.

VADER is a simple rule-based model for general sentiment analysis optimized for social media. It provides four sentiment scores: negative, positive, neutral, and compound (an overall sentiment score from −1 to + 1). VADER is particularly effective at handling sentiments expressed in short, informal text and accounts for factors like punctuation, capitalization, and modifiers (e.g., intensifiers like "very") that influence the intensity of the sentiment. Whereas LIWC excels in offering detailed psychological and linguistic insights across a wide range of texts, VADER is more attuned to detecting sentiment polarity and intensity in social media. Finally, text lengths were extracted as features for each question and included in the models.

Open Vocabulary. LDA is a probabilistic clustering method that identifies topics based on word co-occurrence rather than relying on predefined dictionaries (Blei et al., 2003; Griffiths et al., 2007). This approach allows LDA to group semantically related words while considering context, reducing word sense ambiguity. The latent topics were extracted from the preprocessed text corpus using probabilistic modeling with the R package *topicmodels* (Grün & Hornik, 2011). In our study, we implemented LDA using two distinct approaches for nomothetic and idiographic analyses. For the nomothetic model, we extracted a uniform set of topics across the entire sample. For the idiographic model, we applied LDA separately for each individual, identifying topics based on their unique word usage patterns. To interpret the extracted nomothetic topics, Table 1 presents the 15 most frequent words per topic, offering a semantic representation of the clustering. Details on topic selection, model tuning, and evaluation metrics are provided in the online supplemental materials Section S1 and Figure S1. Furthermore, Figure S2 depicts the distribution of LDA-derived topics across participants, highlighting variations in topic prevalence within the data set.

GPT: LLM-as-Rater Approach. To generate emotion ratings using LLMs, we used GPT, a deep learning-based AI language model developed by OpenAI. GPT is built on a transformer-based architecture, a neural network design that excels at processing sequential data by using self-attention mechanisms to capture contextual relationships across words. This allows GPT to generate human-like text by predicting the next word in a sequence based on the provided context. Specifically, we used GPT-4o, an advanced version pretrained on a vast data set (45TB), enabling it to generate coherent sentences and perform various tasks such as writing, answering questions, and engaging in conversations. In this study, we implemented GPT-4o as an LLM-as-a-rater, a measurement strategy in which the LLM is provided with a structured prompt specifying an explicit rating task and applies that instructed measurement rule to the text. GPT-4o was applied here as a general-purpose rater without any person-specific adaptation. Recent work has begun developing personalized LLMs that tailor outputs to individual users through fine-tuning or retrieval

Table 1
The 15 Most Frequent Words Within the Six Topics Extracted Using LDA

Topic number	Topic interpretation	Most frequent words
1	Social and evening activities: focused on social interactions and nighttime activities with friends	Friend, talk, sleep, last, night, last_night, see, time, morning, hang, wake, late, talk_friend, boyfriend, hang_friend
2	Academic life: centered on school-related activities and academic responsibilities	Homework, school, think, class, finish, test, now, take, tomorrow, math, studi, right, finals, essay, stress
3	Home activities: capturing leisure activities, particularly around media consumption and family time	Watch, nothing, eat, dinner, play, game, show, family, movie, tv, ate, video, eat_dinner, watch_tv, favorite, watch_movie
4	Family interactions: reflecting family relationships and dynamics	Mom, think, fun, sister, new, dad, brother, room, read, make, made, book, something, fight, clean
5	Daily activities: representing routine daily activities like meals	Go, went, home, walk, outside, lunch, food, drive, music, back, listen, shop, car, hurt, around, dog
6	Personal states and obligations: describing emotional states, needs, and work-related responsibilities	Get, work, feel, want, today, day, need, good, think, just, felt, like, done, tired, sick

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augmentation (Liu et al., 2025). Such approaches were not employed in the current study. We prompted GPT-4o through the application programming interface (API) using Python `openai` package to rate the extent to which a participant experienced one of the following emotions: sadness, anger, and nervousness, on a scale of 1–5 (similar to the scale used in EMA), based on responses to three open-text questions. Using the API ensured application of the same standardized prompt, fixed model parameters (e.g., temperature), and consistent, reproducible outputs across the data set. See Section S2 in the online supplemental materials for the full prompt used to generate GPT responses.

Data Analysis

Model Specification

We began our analysis by extracting text data from EMA responses and conducting both lexicon-based and transformer-based analyses. Next, we applied preprocessing steps (e.g., text normalization, stopword removal, and stemming) to refine the text before performing topic modeling. Finally, we trained machine learning models to predict negative emotions based on the derived text features, evaluated model performance, and determined feature importance. Figure 1 illustrates this analysis pipeline in detail.

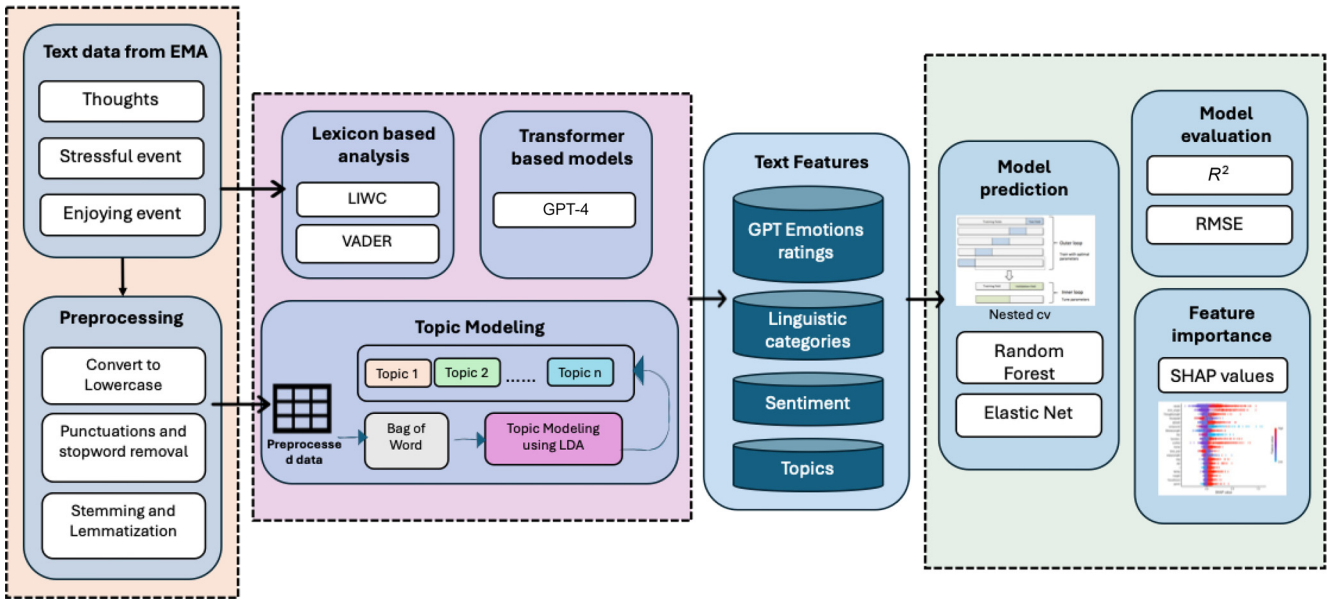
We compared both nomothetic (group-level) and idiographic (individual-level) models in their ability to predict variability in negative emotional states *within* individuals over time. We employed two machine learning approaches to predict negative emotions: Elastic Net Regression (ENR) and Random Forest (RF) models. ENR is a popular variant of conventional regression that combines two types of penalties: ridge and lasso. This combination helps address issues related to multicollinearity by constraining the coefficients of correlated variables while also minimizing model overfitting. On the other hand, RF is an ensemble learning method based on decision trees. Unlike ENR, RF can capture complex nonlinear relationships and interactions between variables without having to specify them in advance. Given that these two methods rely on different algorithms (i.e., penalized regression vs. decision tree-based approaches) for selecting variables, assessing their importance, and generating predictions, comparing them can help to determine which one provides the most accurate predictions in a given context.

To assess model performance and generalizability while minimizing overfitting, we implemented a nested cross-validation (CV) procedure using the nestedcv package in R (Lewis et al., 2023). A full description of the nested CV procedure is provided in Section S3 in the online supplemental materials. We evaluated model accuracy using R^2 (the square of the correlation coefficient) and root-mean-square error (RMSE). R^2 was our primary metric because we wanted to prioritize the model's ability to detect increases in negative emotional states, reflected in corresponding increases in predicted scores. Although a stronger correlation (and thus higher R^2) indicates better alignment between predicted and observed values, it does not necessarily imply lower average prediction error, which is captured by the RMSE.

Initially, we ran nomothetic models using features extracted from all three NLP approaches, including all subjects for group-level

Figure 1

The Pipeline for Processing and Analyzing Text Data From EMA Responses to Predict Emotional States



Note. The process begins with preprocessing (e.g., text normalization, stopword removal, and stemming). Text features are then extracted using lexicon-based analysis (LIWC and VADER), transformer models (GPT-4o), and topic modeling (LDA). These features, including GPT emotion ratings, linguistic categories, sentiment scores, and topics, are fed into machine learning models (RF, Elastic Net) for prediction. Model performance is evaluated using R^2 and RMSE, although SHAP values provide insight into feature importance. This approach integrates multiple NLP techniques to enhance emotion prediction accuracy. EMA = ecological momentary assessment; LIWC = Linguistic Inquiry and Word Count; VADER = Valence Aware Dictionary and sEntiment Reasoner; LDA = Latent Dirichlet Allocation; GPT = Generative Pre-Trained Transformer; RMSE = root-mean-square error; NLP = natural language processing; SHAP = SHapley Additive exPlanations. See the online article for the color version of this figure.

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analysis. These models identified common predictors of negative emotional states at the group level. To evaluate how well these group-level models generalized to individual participants, we calculated individual metrics, such as R^2, derived from the nomothetic models for each participant. Subsequently, we employed fully idiographic models, building separate models for each participant to capture person-specific language-emotion associations. These models focused exclusively on within-person variability, aiming for highly individualized predictions. In these idiographic models, variables with low variance were removed based on standard frequency criteria, where the most common value for that variable could not exceed 95% of the total observations. Finally, we compared the three abovementioned approaches, to understand their relative strengths in predicting negative emotions.

NLP feature extraction procedures were consistent across modeling approaches, with the exception of LDA topics, which were estimated at the sample level for the nomothetic analyses and separately for each participant at the idiographic level. GPT ratings and LIWC/VADER features were identical across approaches.

Feature Importance

To evaluate feature importance, we used SHAP (SHapley Additive exPlanations) values, which quantify the contribution of each feature to the model's predictions in a consistent and interpretable manner by assessing how variations in a feature impact the model's output (Lundberg, 2017). SHAP values were calculated using the R packages fastshap and ggbeeswarm, which facilitates visualization and interpretation of feature contributions. This method allowed us to identify the relative importance of predictors in the model and their specific effects on the predicted outcomes.

Transparency and Openness

Because the data involve sensitive information, and in accordance with IRB requirements, they are available upon reasonable request to the corresponding author (HF) and contingent on a data use agreement. All analysis code and raw output files are available at https://osf.io/grjy9 (Fisher, 2026). Model results were saved separately for each participant, model type, and emotion, resulting in multiple CSV files organized by model type (nomothetic, idiographic, NLP approaches, and ML models). The aggregated values reported in the article (e.g., mean R^2 across participants) were derived directly from these raw outputs using the provided analysis code. We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. This study was not preregistered.

Results

The full results can be explored in the web application (https://emotracknlp.streamlit.app/), which also includes an interactive chatbot powered by a GPT-based LLM. The chatbot was trained on the full article and is designed to answer questions about the research, figures, and results presented in this article, as well as additional findings.

Demographic and Clinical Characteristics

The original data set included 156 participants. Of these, 57 had fewer than 30 EMA observations and were excluded, leaving 99 participants. Two additional participants showed no within-person variation in their EMA emotion ratings across the entire study and were therefore removed, resulting in a base analytic sample of 97 participants. For specific emotion analyses, additional exclusions could occur when EMA sessions were removed through listwise deletion due to missing feature values (e.g., LIWC/VADER, LDA topics, or GPT-derived ratings). In some cases, nonzero emotion ratings occurred only in sessions with missing feature values; after listwise deletion, the remaining sessions therefore contained only zero ratings, making model performance undefined for that emotion. These participants were excluded from that specific emotion analysis. In addition, idiographic models used 10-fold cross-validation within each participant. For participants with very few nonzero observations, some test folds could contain only zero ratings, making performance undefined and leading to additional exclusions from idiographic results. As a result, idiographic sample sizes were sometimes smaller than nomothetic ones. Table 2 presents the demographic and clinical characteristics of the participants. For the participants who completed EMA in four consecutive weeks (see the Method section for EMA schedule), the median number of EMA surveys completed was 77 ($M = 75$, $SD = 24$), and mean EMA compliance was 67% ($SD = 21\%$). For the participants who completed EMA in biweekly 5-day blocks, the median number of EMA surveys completed was

Table 2
Demographic and Clinical Characteristics of the Sample ($N = 97$)

Sample characteristic	*n*	%
Biological sex		
Female	75	77.3
Male	22	22.7
Race		
American Indian	0	0
Asian	13	13.4
Black	10	10.3
Pacific Islander	1	1.0
White	65	67.0
Multiracial	5	5.2
Other race	3	3.1
Ethnicity		
Hispanic	4	4.1
Not Hispanic	92	94.8
Unknown	1	1.0
Current diagnoses (*DSM-5*)		
MDD	21	21.6
GAD	17	17.5
SAD	5	5.2
Panic disorder	3	3.1
Specific phobia	2	2.1
ADHD	1	1.0
PTSD	3	3.1
Medication		
Psychotropic medication	8	8.2

Sample characteristic	*M* (Range)	*SD*
Age (years)	16.2 (12–18)	1.9
Family income ($)	198,895 (0–500,000)	111,478

Note. MDD = Major Depressive Disorder; GAD = Generalized Anxiety Disorder; SAD = Social Anxiety Disorder; PTSD = Posttraumatic Stress Disorder; ADHD = Attention Deficit/Hyperactivity Disorder; *DSM-5 = Diagnostic and Statistical Manual of Mental Disorders*, fifth edition.

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66 (M = 68, SD = 24), and mean EMA compliance was 51% (SD = 17%).

Model Performance: Nomothetic Approach

Table 3 presents model performance metrics for predicting negative emotions using combined text features from three NLP techniques. Three modeling approaches are compared: group-level nomothetic (one model combining all observations), per-participant nomothetic (same modeling approach as group-level nomothetic but performance is calculated separately for each individual), and idiographic (separate models for each individual). Further details and results, including separate models for each NLP technique, are available in the web application (see "model performance across sample" and "model performance per participant").

The group-level nomothetic model performed well, with R^2 values ranging from .11 to .38. However, when evaluated separately for each participant—which is critical given the ultimate goal of tracking fluctuations in emotions *within* individuals—performance decreased substantially. Mean R^2 values fell between .06 and .11, with considerable variability in predictive accuracy across participants. For example, when predicting mean NA, individual R^2 values ranged from .00 to .41, demonstrating that although text features explained little to no variability in NA for some, they accounted for a significant proportion of variability in others. See Table 3 for results for the specific emotions of sadness, anger, and nervousness.

Model Performance: Idiographic Approach

The idiographic approach (separate models for each individual) showed similar average performance to the nomothetic approach, with mean R^2 values ranging from .06 to .10. However, the idiographic approach demonstrated better performance in terms of RMSE (.47–.76) compared with the nomothetic model (.52–.88), suggesting lower prediction error. Figure 2 displays the relationship between predicted and observed (actual) NA ratings using person-specific (idiographic) models. Each colored line represents an individual participant's predicted estimates (y-axis) across different levels of actual NA (x-axis). Although the overall trend, represented by the dashed black line, suggests a generally positive relationship between predicted and actual ratings, indicating that the models capture meaningful within-person fluctuations in NA, there were large between-person differences in the strength of these associations.

Figure 3 displays the substantial differences in model accuracy across individuals by showcasing examples of four high- versus low-accuracy person-specific models (figures for all other outcomes and for all participants for both the nomothetic and idiographic models are available on the web application "True vs. Predicted").

Examining Between-Person Differences in Which Text Features Predict Negative Emotions

Figure 4 illustrates the variability in feature importance by showing the top 10 text features from the four best-performing subject-specific RF models predicting NA. These models are the same four high-accuracy models displayed in Figure 3. The results show two types of variability between individuals: differences in model performance and, even among highly accurate models, differences in the text features that contribute to predicting NA. Critically, this visualization demonstrates how text features provide valuable contextual information. For example, one participant (Participant 2) exhibits more negative emotions when using words related to their family ("Family"), yet another (Participant 3) shows greater

Table 3

Comparison of Group-Level Nomothetic; Per-Participant Nomothetic; and Idiographic Model Performance for Negative Affect, Sadness, Anger, and Nervousness Using Random Forest and Elastic Net

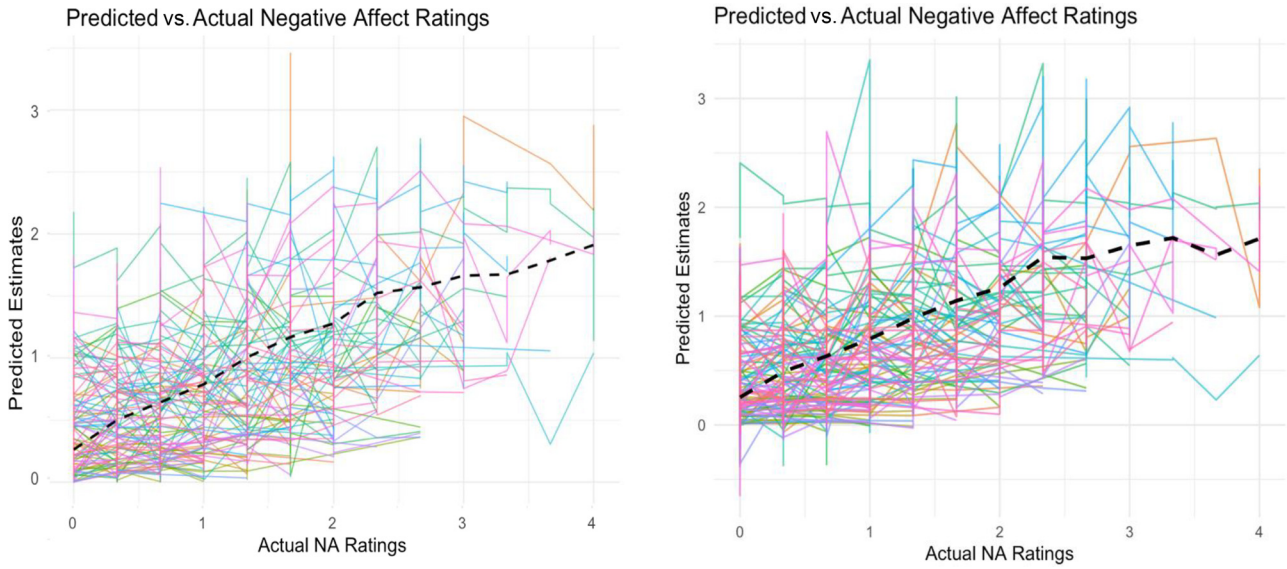
Emotion outcome and type of model	Nomothetic group-level performance			Nomothetic per-participant performance				Idiographic			
	R^2	R	RMSE	R^2	R^2 range	R	RMSE	R^2	R^2 range	R	RMSE
Negative affect											
Random Forest	.38	.62	0.55	.10 (.10)	.00; .41 Sig. N = 53/97	.26 (.17)	0.52 (0.22)	.10 (.12)	.00; .47 Sig. N = 42/97	.21 (.25)	0.47 (0.20)
Elastic Net	.17	.41	0.64	.11 (.09)	.00; .39 Sig. N = 56/97	.28 (.17)	0.61 (0.27)	.10 (.12)	.00; .53 Sig. N = 47/97	.15 (.28)	0.48 (0.20)
Sad											
Random Forest	.33	.58	0.75	.10 (.12)	.00; 0.53 Sig. N = 48/96	.25 (.21)	0.73 (0.27)	.10 (.13)	.00; .46 Sig. N = 38/94	.17 (.26)	0.68 (0.22)
Elastic Net	.14	.37	0.85	.09 (.09)	.00; .42 Sig. N = 51/96	.24 (.19)	0.82 (0.32)	.10 (.11)	.00; .51 Sig. N = 46/94	.10 (.30)	0.69 (0.23)
Angry											
Random Forest	.20	.45	.69	.09 (.12)	.00; .69 Sig. N = 37/90	.22 (.20)	0.63 (0.33)	.06 (.07)	.00; .33 Sig. N = 27/85	.08 (.23)	0.64 (0.29)
Elastic Net	.13	.36	.72	.07 (.07)	.00; .45 Sig. N = 49/90	.19 (.10)	0.88 (0.32)	.07 (.08)	.00; .43 Sig. N = 39/85	.09 (.27)	0.66 (0.31)
Nervous											
Random Forest	.24	.49	.85	.06 (.08)	.00; .44 Sig. N = 33/95	.17 (.06)	0.81 (0.29)	.08 (.12)	.00; 0.71 Sig. N = 33/94	.10 (.27)	0.75 (0.27)
Elastic Net	.11	.33	.92	.07 (.07)	.00; .35 Sig. N = 42/95	.18 (.07)	0.88 (0.33)	.09 (.12)	.00; 0.84 Sig. N = 43/94	.12 (.30)	0.76 (0.29)

Note. Sig. N represents the number of participants for whom the association between actual ratings and model-predicted ratings was statistically significant. RMSE = root-mean-square error.

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Figure 2

The Relationship Between Predicted and Observed (Actual) Negative Affect Ratings Using Person-Specific (Idiographic) Models

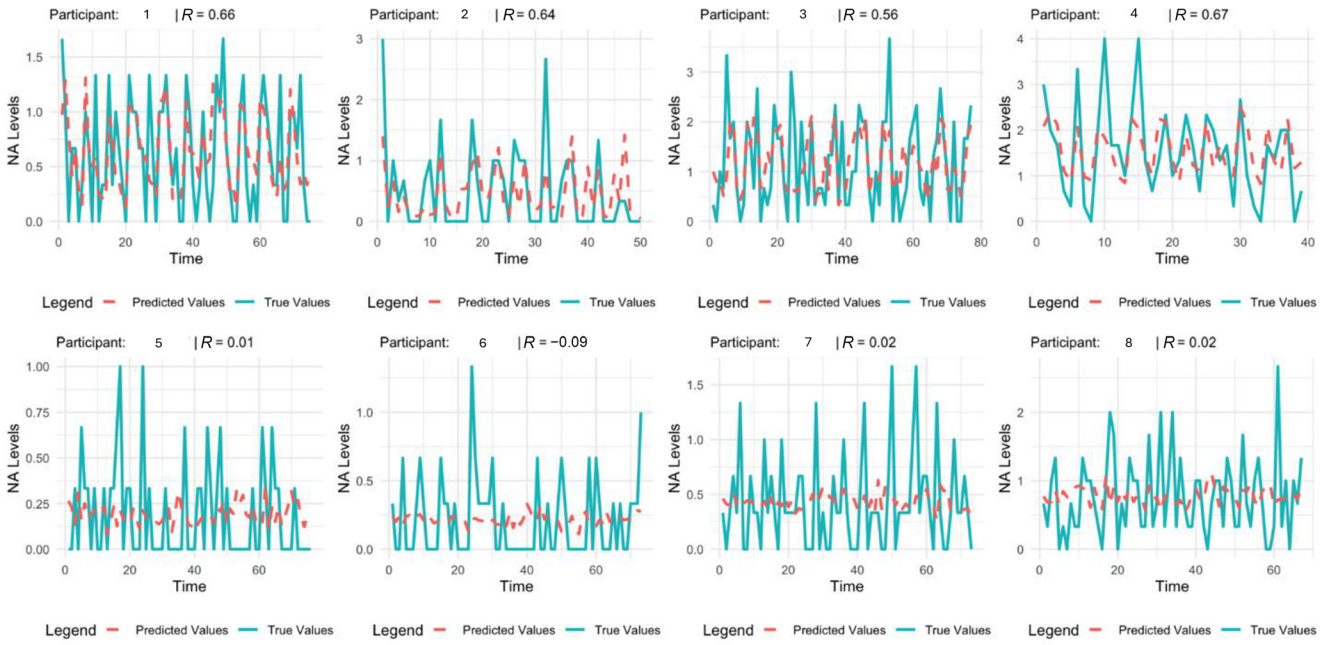


Note. Panel A: Random Forest. Panel B: Elastic Net. Each colored line represents an individual participant's predicted estimates (*y*-axis) across different levels of actual negative affect (*x*-axis). NA = negative affect. See the online article for the color version of this figure.

negative emotions when focusing on the past ("FocusPast"). Similarly, a different participant (Participant 4) expresses more negative emotions when writing about work-related topics ("Work") but demonstrates less negativity when using words associated with acquiring objects, states, or goals that fulfill personal needs ("Acquire"). This rich information can be used to create person-specific profiles that enable us to identify the specific contexts in which negative emotions arise, providing insights that can be used to personalize interventions. A heatmap of the feature importance across participants and feature-importance figures for each participant, outcome, and ML model are available on the web application ("feature importance heatmap" and "feature importance per

Figure 3

Examples of Person-Specific (Idiographic) Predictions of Negative Affect for High Performance (Top Panel) and Low-Performance (Lower Panel) Models

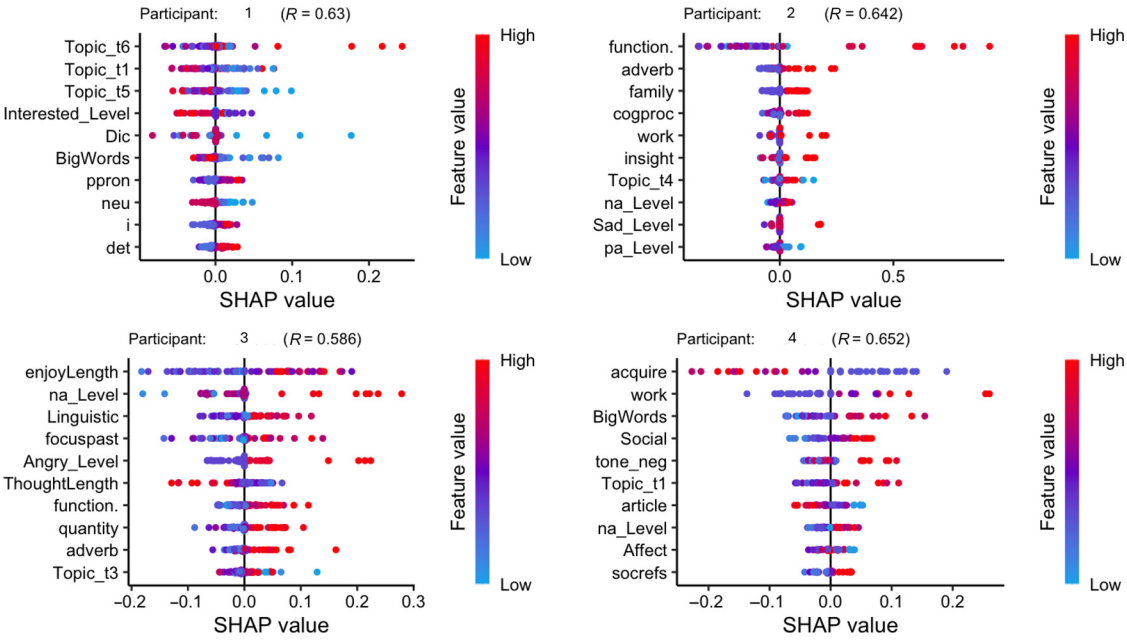


Note. NA = negative affect. See the online article for the color version of this figure.

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Figure 4

SHAP (SHapley Additive exPlanations) Beeswarm Plots for Four Individual Participants, Illustrating the Contribution of Different Text-Based Features to the Model's Emotion Predictions



Note. Each dot represents an individual data point, with red indicating high feature values and blue indicating low feature values. The spread of dots across the *x*-axis reflects the variability in a feature's effect on predictions across different measurement time point. The *y*-axis lists the features in descending order of importance, meaning the top features had the strongest impact on the model's predictions. The *x*-axis represents SHAP values, indicating the magnitude and direction of each feature's impact on the predicted emotion scores. For example, for Participant K23528, a low score on "Acquire" (red color) increases the model's prediction of negative affect. Conversely, a high feature value for "Work" or "Big Words" (red color) contributes to a higher predicted negative affect. See the online article for the color version of this figure.

participant"). Details on the differences in predictive text features between high- and low-performing models can be found in the online supplemental materials (Section S4 and Figure S3).

Which NLP Approach Performs Best in Predicting Negative Emotions?

Table 4 and Figure 5 display the performance metrics for each NLP approach used in the idiographic models. When examining the idiographic predictive performance of each NLP approach separately, GPT generally showed better performance for NA, sadness, anger, and nervousness, with R^2 values around .10 for NA and sadness, and slightly lower for anger and nervousness. These results were comparable to those achieved by the idiographic models that combined all NLP approaches (see Table 2 for the combined model results), except in the prediction of nervousness where GPT slightly outperformed the combined idiographic model. However, it is important to note that although GPT had relatively strong R^2 values, it also had a higher RMSE compared with the combined models, suggesting a higher average prediction error despite accounting for more variance in the outcomes. LIWC+VADER and LDA each demonstrated poorer predictive performance, with the lowest average R^2 values and fewer significant associations across all emotions, especially for anger and nervousness. This highlights that the contextual understanding provided by GPT was more effective in capturing within-person emotional fluctuations compared with the other methods when NLP approaches were tested individually. Overall, although GPT captures variability in negative emotions (as reflected by R^2), its predictions are less precise in terms of exact values (as indicated by RMSE). Combined models, which balance capturing variability and minimizing prediction errors, offer a more robust and reliable approach to emotion prediction.

Evaluating Best-Performing Models for Each Individual

To investigate the effectiveness of personalized models, we identified the best-performing predictive model for each of the 97 participants. Using this flexible approach, the vast majority (90.7%–94.8%) of the participants showed a significant correlation between predicted and observed (actual) emotion scores. The table detailing the best model for each participant and each emotion outcome can be found in the web application under "Best model performance." Figure 6 shows the distribution of best-performing models across participants (see also in the web application under "Best Model Performance"). Idiographic models were most frequently selected for all outcomes except for anger, where nomothetic models performed best for 44.3% of the participants. Elastic Net showed the best performance for most participants across outcomes, followed by RF. For NLP approaches, the model combining all text features outperformed for most of the participants (52.6%–59.8%). Interestingly, though GPT showed the highest mean R^2 scores, it was chosen as the best model for only a relatively small portion of the participants (11.3%–19.6%).

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Table 4

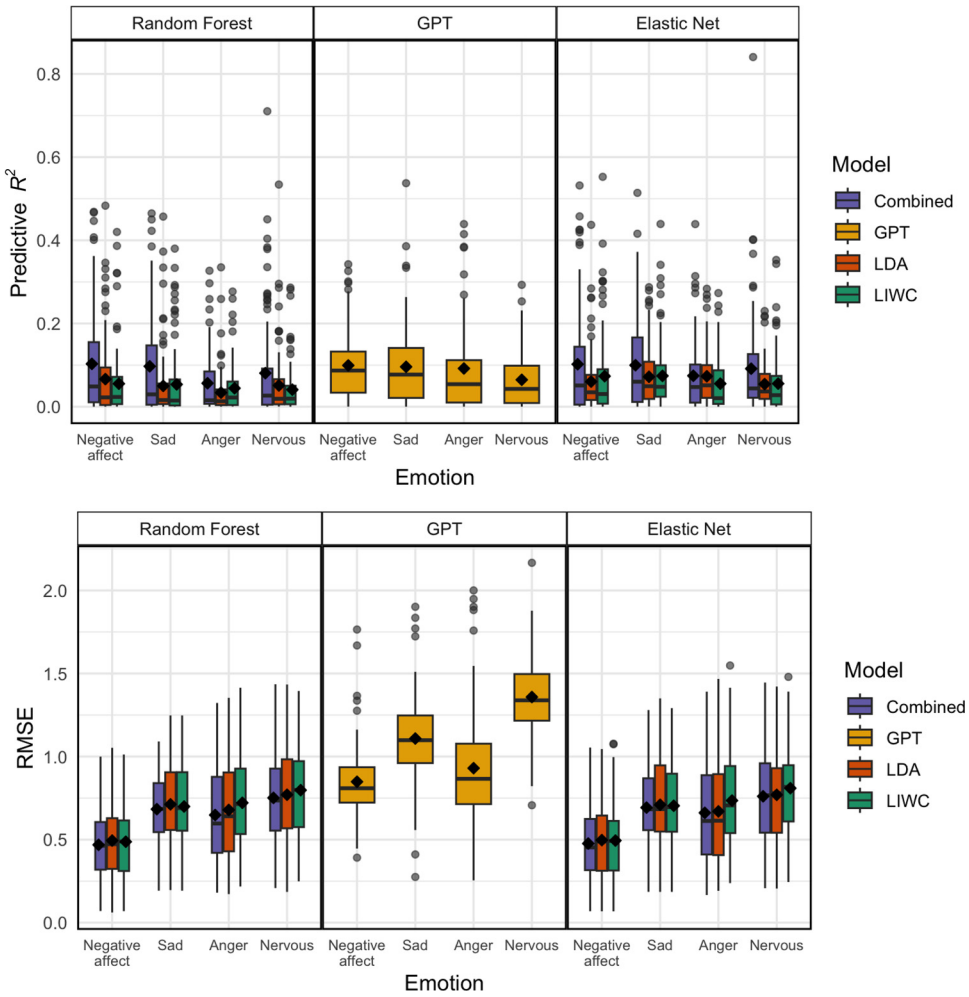
Comparison of Idiographic Model Performance for Negative Affect, Sadness, Anger, and Nervousness Using LIWC + VADER, LDA, and GPT

	LIWC + VADER				LDA				GPT			
	R^2	R^2 range	R	RMSE	R^2	R^2 range	R	RMSE	R^2	R^2 range	R	RMSE
Negative affect												
Random Forest	.05 (.08)	.00; .42 Sig. N = 31/97	.19 (.24)	0.49 (0.21)	.07 (.09)	.00; .48 Sig. N = 36/97	.11 (.23)	0.49 (0.21)	.10 (.08)	.00; .34 Sig. N = 55/97	.26 (.18)	0.85 (0.20)
Elastic Net	.07 (.10)	.00; .55 Sig. N = 36/97	.08 (.30)	0.49 (0.22)	.06 (.07)	.00; .44 Sig. N = 35/97	−.12 (.21)	0.50 (0.22)				
Sad												
Random Forest	.05 (.08)	.00; .38 Sig. N = 24/95	.15 (.25)	0.70 (0.25)	.05 (.08)	.00; .46 Sig. N = 25/95	.08 (.21)	0.71 (0.26)	.10 (.10)	.00; .54 Sig. N = 55/96	.25 (.19)	1.11 (0.26)
Elastic Net	.07 (.08)	.00; .44 Sig. N = 42/95	.02 (.30)	0.70 (0.26)	.07 (.07)	.00; .29 Sig. N = 45/95	−.17 (.20)	0.71 (0.27)				
Angry												
Random Forest	.04 (.06)	.00; .28 Sig. N = 15/70	.06 (.20)	0.72 (0.29)	.03 (.05)	.00; .58 Sig. N = 12/85	.03 (.18)	0.68 (0.32)	.09 (.10)	.00; .44 Sig. N = 45/90	.22 (.21)	0.93 (0.34)
Elastic Net	.05 (.07)	.00; .27 Sig. N = 26/70	−.06 (.25)	0.73 (0.31)	.07 (.07)	.00; .28 Sig. N = 43/84	−.21 (.17)	0.67 (0.32)				
Nervous												
Random Forest	.04 (0.06)	.00; .29 Sig. N = 19/86	.10 (.24)	0.80 (0.26)	.05 (.08)	.00; .53 Sig. N = 29/94	.10 (.20)	0.77 (0.28)	.06 (.06)	.00; .29 Sig. N = 40/95	.18 (.18)	1.36 (0.22)
Elastic Net	.05 (.07)	.00; .35 Sig. N = 25/86	.00 (.28)	0.81 (0.28)	.05 (.05)	.00; .23 Sig. N = 36/94	−.15 (.17)	0.76 (0.28)				

Note. The process begins with preprocessing (e.g., text normalization, stopword removal, and stemming). Text features are then extracted using lexicon-based analysis (LIWC, VADER), transformer models (GPT-4), and topic modeling (LDA). These features, including GPT emotion ratings, linguistic categories, sentiment scores, and topics, are fed into machine learning models (Random Forest, Elastic Net) for prediction. Model performance is evaluated using R^2 and RMSE, although SHAP values provide insight into feature importance. This approach integrates multiple NLP techniques to enhance emotion prediction accuracy. LIWC = Linguistic Inquiry and Word Count; VADER = Valence Aware Dictionary and sEntiment Reasoner; LDA = Latent Dirichlet Allocation; GPT = Generative Pre-Trained Transformer; RMSE = root-mean-square error; SHAP = SHapley Additive exPlanations.

Figure 5

Performance Metrics (R^2 and RMSE) for Each NLP Approach Used in the Idiographic Models



Note. The top panel displays the predictive R^2 for different models (Random Forest, GPT, and Elastic Net) across four emotion categories: negative affect, sadness, anger, and nervousness. The bottom panel presents the corresponding RMSE values. Each box plot represents the distribution of performance metrics for four different NLP-based feature sets: Combined (purple), GPT (yellow), LDA (orange), and LIWC+ VADER (green). Higher R^2 indicates better predictive accuracy, whereas lower RMSE reflects better model fit. RMSE = root-mean-square error; GPT = Generative Pre-Trained Transformer; NLP = natural language processing; LDA = Latent Dirichlet Allocation; LIWC = Linguistic Inquiry and Word Count; VADER = Valence Aware Dictionary and sEntiment Reasoner. See the online article for the color version of this figure.

Are Prediction Models Emotion-Specific?

To examine whether our models captured emotion-specific signals rather than general NA, we compared the extent to which each emotion-specific model (both RF and Elastic Net) predicted each observed emotion. Specifically, for each observed emotion outcome (NA, sad, angry, and nervous), we calculated R^2 and RMSE between the observed and the predicted emotion, either matching (e.g., predicted and observed sadness) or mismatching (e.g., predicted anger and observed sadness). Figures S4–S7 display box plots of the distribution of R^2 and RMSE for each emotion. As shown in the figures, overall, and consistent with the interpretation that the models have emotion-specific signals, the matched predictions showed higher R^2 and lower RMSE than mismatched predictions, indicating that the models capture unique, emotion-specific variance. However, for RMSE, we sometimes observed better performance for NA relative to the specific emotions, suggesting that the signals underlying broad NA might be more robust or reliable relative to single emotion ratings.

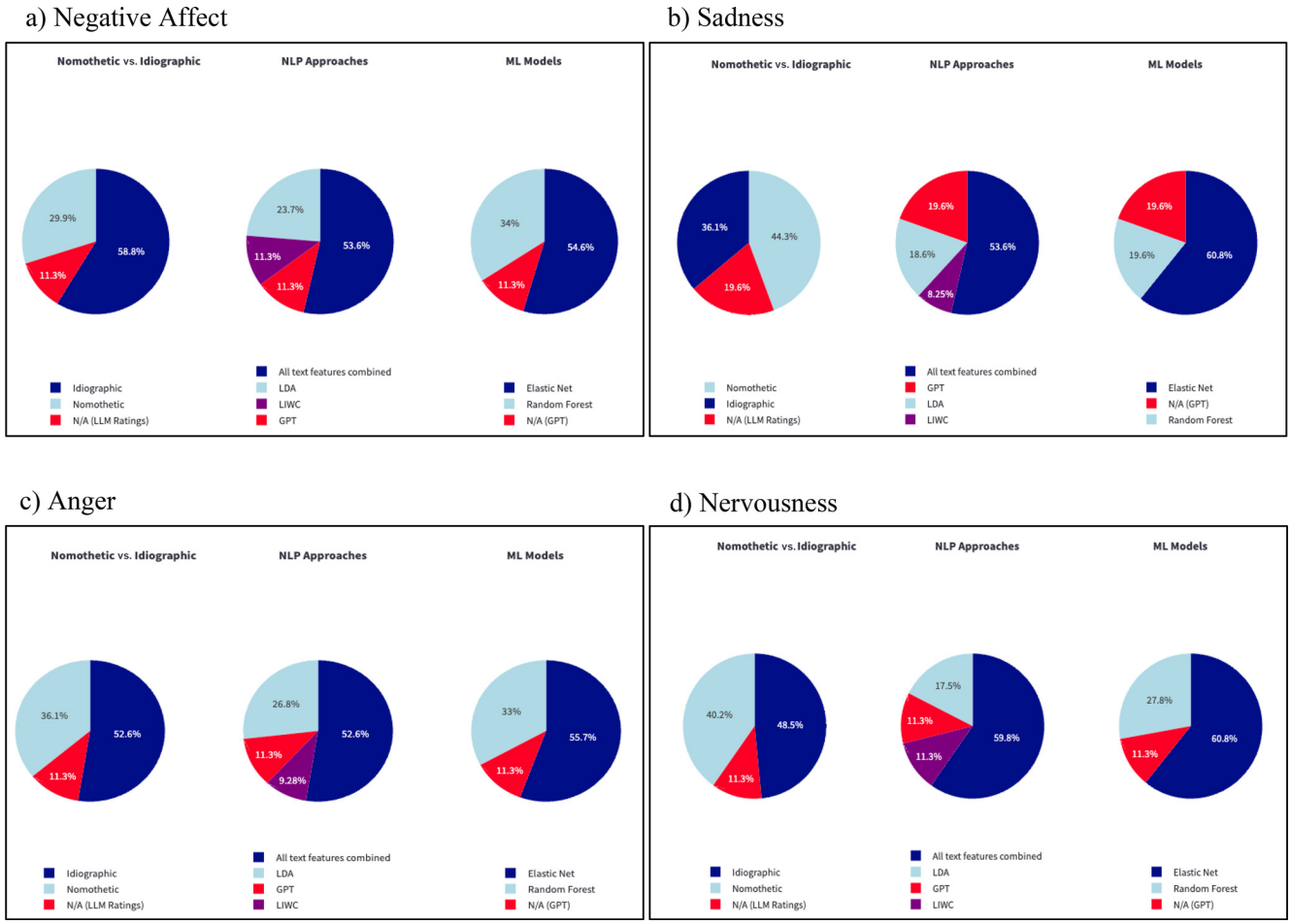
Does Model Performance Vary by Symptom Levels, Demographic Characteristics, or Number of Observations?

We conducted exploratory analyses examining whether model performance varied as a function of symptom levels, demographic characteristics, and number of observations (Tables S1 and S2 in the online supplemental materials). After False

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Figure 6

Distribution of Best-Performing Models for Predicting Negative Emotions Across Nomothetic Versus Idiographic Frameworks, NLP Approaches, and ML Models



Note. The colors represent different model types, indicating the proportion of participants for whom each model was the best predictor. NLP = natural language processing; ML = machine learning; LLM = large language model; LDA = Latent Dirichlet Allocation; GPT = Generative Pre-Trained Transformer; LIWC = Linguistic Inquiry and Word Count; GPT = Generative Pre-Trained Transformer. See the online article for the color version of this figure.

Discovery Rate correction (FDR) correction, only higher anhedonia (SHAPS) was associated with better prediction of sadness in the idiographic RF model ($\rho = .35$, $p = .001$, pFDR = .032). No other associations were significant after correction, including demographic variables and number of observations.

Discussion

In this study, we combined multiple NLP approaches to examine whether within-person fluctuations in self-reported NA can be inferred from EMA narratives using NLP. By comparing different modeling strategies and language-based features, the results provide a benchmark for future research examining language-based inference of affect in EMA contexts. Recognizing the idiosyncratic nature of emotional communication, we compared idiographic models, tailored to individual patterns, with nomothetic models that capture common trends across groups. The use of NLP and machine learning techniques to analyze language may provide new opportunities for studying emotions and the contexts in which they arise in daily life, which could ultimately inform future approaches to mental health assessment and clinical decision-making.

Comparing Nomothetic and Idiographic Models

The results showed that, overall, group-level nomothetic models showed high performance in continuously predicting negative emotions (R^2 range: .11–.38). These findings align with previous studies demonstrating the utility of NLP approaches in detecting emotional states across participants (Akhtar et al., 2022; Tanana et al., 2021). However, as these studies primarily focused on identifying emotional states at the between-person level, rather than capturing *within-person* fluctuations, their high performance does not necessarily indicate an ability to accurately track moment-to-moment emotion fluctuation within individuals. When we calculated performance metrics for each participant individually, the model performance declined significantly and showed high

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between-person variability, indicating that the nomothetic models' ability to track within-person changes varies considerably from one individual to another.

When we built separate models for each participant (idiographic models) and compared their performance to that of nomothetic we found that the mean R^2 was comparable between the two approaches, however, RMSE was lower for the idiographic models. In addition, idiographic models revealed significant between-person variability in the text features predicting emotional states, indicating that individuals express negative emotions in distinct linguistic ways. This raises an intriguing question: If individuals differ so markedly in their predictive features, how can a nomothetic model that captures only general trends perform just as well, at least in terms of R^2? One plausible explanation is a trade-off between statistical power and individual variation. Although idiographic models capture unique, person-specific patterns, nomothetic models benefit from larger data sets that enable the estimation of multiple, weaker yet common feature-emotion relationships. In other words, although individual differences exist, the robust common patterns identified by the nomothetic approach appear sufficient to achieve similar predictive accuracy. Supporting this interpretation, idiographic models showed lower RMSE and higher accuracy for more participants compared with the nomothetic models. However, nomothetic models demonstrated greater consistency across participants, with more participants presenting significant associations between predicted and actual negative emotions compared with idiographic models.

Rather than directly comparing nomothetic and idiographic models, we also tested an approach that selected the best-performing model for each participant. Using this method, idiographic models were most often the best performers, and the vast majority of participants (90.7–94.8%) showed a significant correlation between predicted and observed emotion scores. A limitation of this approach is that choosing the "winner" per participant may lead to overfitting or capitalize on chance, inflating accuracy estimates. At the same time, the results suggest that some participants may need individualized models to capture their unique patterns of expression, whereas others can be modeled well using data pooled across individuals. Balancing personalization and generalizability is therefore a key challenge for NLP and machine learning approaches to emotion prediction.

Although we are not aware of studies directly comparing nomothetic and idiographic approaches in text analysis, previous research comparing their effectiveness in tracking mental states using passive sensor data from smartphones and actigraphy, as well as self-reported EMA, has yielded similar results (Aalbers et al., 2023; Cheung et al., 2017; Rozet et al., 2019; Soyster et al., 2022). For example, Aalbers et al. (2023) used smartphone passive sensor data to predict stress levels. Their findings revealed that idiographic models demonstrated higher accuracy in tracking stress levels for some participants (Spearman's ρ rank-order correlation up to 1 in idiographic models vs. up to .65 in nomothetic models). However, nomothetic models significantly predicted stress for a larger proportion of participants (up to 23.2% for idiographic models vs. up to 55% for nomothetic models). Rozet et al. (2019) used a comparable method and initially found that nomothetic models performed better (i.e., were more accurate) than idiographic models. However, as more data accumulated, the performance of the idiographic model eventually equaled and then surpassed that of the nomothetic model, suggesting that idiographic models may be a better option when sufficient data are available at the single subject level.

Comparing NLP Approaches

When comparing specific NLP approaches, GPT-4o, implemented here as an LLM-as-rater, achieved the highest R^2 among the individual models, comparable to the model combining all three NLP approaches. However, GPT-4o also showed higher RMSE than the combined model, suggesting that despite capturing meaningful variance in emotional states, it was less precise in predicting their exact numerical values. One possible contributor to this pattern is that GPT ratings tended to show lower variance than the EMA ratings, which can lead to larger residuals and therefore higher RMSE. Furthermore, when selecting the best-performing model for each participant, GPT was chosen for only a relatively small portion of individuals, indicating that its predictions are relatively stable across individuals but not necessarily the most accurate for any single participant. This pattern aligns with the nomothetic-idiographic trade-off discussed earlier, as GPT operates as a nomothetic model that generalizes across individuals rather than incorporating person-specific information. Future research should aim to optimize this balance by leveraging fine-tuning techniques, such as participant-specific calibration or supervised fine-tuning on emotion-labeled data, to improve GPT model accuracy.

From an applied perspective, GPT is already trained on a vast corpus of data (approximately 45 terabytes of text data from various sources [Achiam et al., 2023]), which makes it particularly attractive for studies with limited data sets, as it does not necessarily require retraining on task-specific data to yield meaningful predictions. However, the model's lack of transparency and inability to explicitly articulate the rationale behind its predictions remain critical limitations that researchers must carefully weigh (Feuerriegel et al., 2025). Given these considerations, we recommend that researchers thoughtfully evaluate GPT's role in their analytical pipeline, considering whether to employ it as a standalone tool or integrate it with complementary approaches. Future work should focus on developing methods that combine GPT's powerful inference capabilities with more transparent analytical approaches, potentially offering a more robust framework for emotion analysis in psychological research.

Finally, exploratory analyses suggested slightly better prediction among adolescents with higher symptom levels, although these associations were weak and no consistent effects were observed for demographic characteristics. If replicated in larger samples, this pattern may further support the potential relevance of language-based emotion prediction for identifying early markers of affective disorders.

Limitation and Future Directions

This study is the first to use different NLP approaches to directly compare nomothetic and idiographic models for tracking emotional fluctuations. Yet findings of this study should be interpreted in light of several limitations. First, capturing the nuanced dynamics of emotional expression typically requires large amounts of data. In everyday interactions, subtle emotional changes may only become apparent with extensive exposure to an individual's language use, much like knowing someone well enables you to discern small shifts in their mood. In our analyses, we did not find a significant association between the number of observations and predictive performance. This suggests that, within the range of

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observations available in the current data set, differences in sample size were unlikely to account for the observed differences in accuracy. However, it remains possible that substantially larger numbers of observations per participant could improve idiographic model performance, and future studies with longer monitoring periods may be better positioned to evaluate this possibility. Therefore, future studies should strive to collect larger within-person data sets, potentially sourced from daily-life communications, to enhance the ability to track and model these nuances. Second, only one generative LLM was evaluated, and performance may vary across different models or model versions.

Third, the overall predictive accuracy was modest, leaving significant room for improvement. In addition to increasing the quantity of data, future research could benefit from incorporating additional modalities (e.g., vocal features, facial expression, passive sensors from smartphones or wearables such as smartwatches or smart rings, which are becoming increasingly popular) alongside text. Prior studies have shown that combining various modalities can enhance the performance of nomothetic models of emotion (Kalateh et al., 2024), and thus this multimodal approach may also improve idiographic models. Moreover, future work should explore whether individuals not only differ in the text features that predict their emotions but also in the types of modalities that best capture their emotional states. For example, individual differences, such as the tendency to use affective suppression, may influence how emotions are conveyed in text, with greater suppression potentially leading to lower model performance. Future studies should examine whether these participants require larger data sets to capture subtle nuances in their text, or if alternative modalities (e.g., passive sensor) better detect changes in their emotional states.

Fourth, the current study compared idiographic and nomothetic approaches, demonstrating that each has its merits. Future studies could explore hybrid approaches that integrate both individual-specific and group-level information, balancing personalization with statistical power to enhance predictive accuracy. Finally, the text data in our study consisted of responses to specific questions, which limits the generalizability of our findings to other types of text (e.g., text messages and social media posts). Future research should incorporate text from diverse sources such as social media, conversational exchanges, and free-form writing, to determine whether these results extend to broader contexts.

Finally, in this study, NA was reported before the narrative prompts, reducing the likelihood that recalling and describing emotionally valenced events inflated concurrent affect ratings. However, because two prompts explicitly asked participants to describe stressful or positive experiences, predictive performance likely reflects, at least in part, the emotional focus of the prompts. This is consistent with recent meta-analytic work on depression prediction showing that text type matters, with depression-relevant text (e.g., clinical interviews) yielding stronger predictive performance than more general language in predicting depression (Fisher et al., 2026). Accordingly, the findings should be interpreted within the context of this structured EMA protocol.

More broadly, written EMA narratives likely capture participants' appraisal and narrative framing of emotionally salient events rather than serving as direct readouts of internal emotional states. In this sense, the present study evaluates whether self-reported NA can be inferred from these constructed narrative accounts within a structured EMA context. Although such narratives may reflect how adolescents describe positive and negative experiences in their daily lives, the findings may not generalize to spontaneous or unprompted language use, or to other communication modalities, such as spoken language, where affective signals are conveyed through features like prosody and timing.

Conclusion

In conclusion, our findings highlight the potential of combining NLP approaches for modeling within-person variation in self-reported NA from EMA narratives, demonstrating both the strengths and limitations of idiographic and nomothetic models. Nomothetic models effectively capture general trends, making them useful for broad applications. In contrast, idiographic models provide a more nuanced understanding by identifying person-specific features that reflect an individual's unique emotional expression, though they require large within-person data sets to model these patterns reliably. Rather than a one-size-fits-all approach, our results suggest that selecting the best approach for each participant can enhance predictive accuracy. Expanding this work to incorporate diverse text sources and multimodal data streams (e.g., integrating social media posts, speech transcripts, facial expressions, physiological signals, or smartphone sensor data) may further advance the field.

Ultimately, improving methods for inferring emotions and their contextual drivers from text may enhance our ability to study emotions in daily life and inform the development of interventions that consider not only moments of distress but also the specific emotions and contexts in which they arise, enabling more personalized support for youth.

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Received October 5, 2025
Revision received March 19, 2026
Accepted May 13, 2026 ■

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